

A Synopsis on
An AI-Powered College FAQ Chatbot

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I. Introduction to Problem

There has been a growing demand for educational institutions to provide students with accurate and timely answers to common questions. Questions about admissions, exams, timetables, payment, facilities, etc., are commonly asked by students and the responses to such questions are time-consuming to produce, create additional work for staff, and lead to inconsistency in response delivery to students.

Conventional chat systems that were designed based on rules are often not capable of comprehending queries posed in natural language, nor do they have the capacity to offer contextual responses.

Though, through recent advancement in LLMs, RAGs and other methods, intelligent conversational systems can now be developed that collect data from institutions' knowledge bases and provide accurate and timely responses to students and staff. RAG provides the added security of providing greater response reliability through combining document retrieval integrity with generative models; therefore, reducing hallucination and providing a better overall understanding of the context of the question being asked.

This Document will outline the design and creation of an AI-based college FAQ Chatbot that utilizes LLM API's, a vector-based retrieval system and an analytic dashboard. The chatbot will have access to an institution's current and previously developed FAQ datasets and can respond to student inquiries, as well as provide administrators with usage data and analytics through graphical representation of data. The focus of the project is to improve best practice in developing student service through automation and efficient communication among institutions.

II. Literature Survey

A variety of technologies currently exist which serve as basis for commercial chatbots, including rules-based, traditional Artificial Intelligence (AI) (machine-learning) and generative AI-based systems as well as retrieval-augmented-generative artificial intelligence (RAG) or Intelligent Assistants (IA) that utilize RAG. Rules-based systems are able to provide predictable answers to user queries; however, they do not offer scalability. One downside of ML chatbots is that in order for them to learn, they must have access to large quantities of data; hence, the more data they have, the better able they will be to provide accurate answers to user queries. LLM chatbots offer the user natural conversation; however, when it comes to generating responses outside of their training data, LLM will give the user inaccurate information because LLM have no way of retrieving external data when generating their response. RAG systems allow for the retrieval of additional documents from which to generate their response to user's question thus making RAG-based AI systems superior to traditional AI systems for automating FAQ systems. RAG-based AI systems have also shown statistically significant improvements in terms of accuracy and contextual relevance when compared with traditional AI chatbot architecture in studies conducted through July 2023.

III. Comparative Study

System	Technology Used	Advantages	Limitations
Rule-Based Chatbot	Decision rules, predefined responses	Easy implementation	Cannot handle complex queries
Machine Learning Chatbot	NLP classification models	Learns from data	Requires large datasets
Generative AI Chatbot	Large Language Models	Natural conversations	Possible hallucinated answers
Retrieval-Based Chatbot	Similarity search databases	Accurate domain responses	Limited flexibility
RAG-Based Chatbot	LLM + vector database retrieval	Context-aware, high accuracy	Complex architecture
API-Integrated Chatbot Platforms	Cloud LLM APIs + web interfaces	Easy deployment	Dependency on external services

IV. Objective

1. Build a Chatbot that responds automatically to College frequently asked questions (FAQs) and uses Augmented Document Generation (ADG) as a basis for their contextual relevance.
2. Incorporate an API to allow the use of Large Language Models (LLMs) for an interactive conversation interface.
3. Create a Data Visualization Tool (DVT) that measures and analyzes the use of the Chatbot.
4. Create an API Driven scalable Chatbot system for implementation.

V. Planning of work:

1. Data Collection:

Collect institutional FAQ datasets including admission details, fee structures, course information, examination schedules, and campus-related queries from official college sources and documents.

2. Data Preprocessing and Embedding Generation:

Clean and preprocess the collected textual data by removing noise, formatting inconsistencies, and duplicate entries. Convert the processed text into numerical vector embeddings using NLP embedding models such as Sentence Transformers.

3. **Vector Database Storage:**
Store the generated embeddings in a vector database such as FAISS or ChromaDB to enable efficient similarity-based search and fast retrieval of relevant information.
4. **Retrieval Pipeline Development:**
Implement a retrieval mechanism that converts user queries into embeddings and searches the vector database to identify the most relevant FAQ entries based on similarity scores.
5. **LLM Integration:**
Integrate a Large Language Model (LLM) API to generate context-aware responses using the retrieved documents as input, enabling accurate and conversational chatbot replies.
6. **Backend Development:**
Develop the chatbot backend using Python frameworks such as FastAPI or Flask to create API endpoints that process user queries and return chatbot responses.
7. **Visualization Dashboard:**
Build an interactive dashboard using Streamlit or similar visualization tools to monitor chatbot usage statistics, query frequency, and response performance analytics.
8. **Testing, Optimization, and Deployment:**
Conduct system testing for performance, accuracy, and response time. Optimize retrieval and response generation modules and deploy the final chatbot system on a cloud-based hosting platform for real-time usage.

VI. Bibliography/References

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